KELLY N. SANDERSON

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SUMMARY

I am currently a Postdoctoral Researcher in Extragalactic Astrophysics at Haverford College. In this role, I am assisting both the Roman Science Collaboration and the Galaxy Zoo collaboration to set up the first Roman Galaxy Zoo project! I earned my PhD in Astronomy from New Mexico State University. I am passionate about increasing both the accessibility and approachability of astronomy to underrepresented groups through mentoring, outreach, and a better understanding of students' mindsets about learning.

RESEARCH INTERESTS

I am interested in the timescales over which AGN impart feedback on their host galaxies and environment, an area known as *AGN Archaeology*, as well as the cosmic co-evolution of AGN and their host galaxies. My research uses *radio*, *infrared*, and *optical* observations to identify AGN, constrain the timescales associated with their different feedback (accretion) modes, and study the properties of their host galaxies.

EDUCATION

New Mexico State University, Las Cruces, NM — *August 2018 - May 2025*
PhD in Astronomy — Cumulative GPA: 3.84/4.0

New Mexico State University, Las Cruces, NM — *August 2018 - May 2023*
MS in Astronomy — Cumulative GPA: 3.84/4.0

Old Dominion University, Norfolk, VA — *August 2013 - May 2018*
BS in Physics — Cumulative GPA: 3.28/4.0
Minor in Mathematics

EMPLOYMENT

Haverford College
Haverford, PA
Postdoctoral Researcher in Extragalactic Astrophysics — *April 2026-current*

New Mexico State University
Las Cruces, NM
Research Associate — *June 2025-March 2026*
Graduate Research Assistant — *May 2019-May 2025*
Teaching Assistant (*online*) — *August 2021-May 2022*
Teaching Assistant — *August 2018-May 2019*

National Radio Astronomy Observatory
Socorro, NM
Graduate Summer Research Assistant — *June 2021-August 2021*
Graduate Summer Research Assistant — *July 2019-August 2019*

National Radio Astronomy Observatory
Charlottesville, VA
Undergraduate Summer Research Assistant (NAC) — *May 2018-August 2018*

University of Wisconsin
Madison, WI
Undergraduate Summer Research Assistant (NAC) *May 2017-August 2017*

Old Dominion University
Pretlow Planetarium, Norfolk, VA *May 2014-August 2018*
Director's Assistant

GRANTS & AWARDS

2024
NRAO Grote Reber Doctoral Fellowship – extension ($32,000/year) *September 2024-May 2025*

2023
American Astronomical Society International Travel Grant ($2300) *June 2023*
NMSU Graduate School Merit Award Scholarship ($5000) *May 2023*

2022
NRAO Grote Reber Doctoral Fellowship ($32,000/year) *August 2022-August 2024*

2021
NMSU Astronomy Department Zia Reseach Award ($717) *December 2021*
New Mexico Space Grant Consortium (NMSGC) Fellowship ($10,000) *August 2021*
NMSU Outstanding Graduate Assistantship Award ($2000) *May 2021*

2020
NMSU Astronomy Murrell Endowed Memorial Scholarship ($555) *December 2020*
New Mexico Space Grant Consortium (NMSGC) Fellowship ($10,000) *August 2020*

2019
NMSU Astronomy William Webber Voyager Graduate Scholarship ($7500) *August 2019*
National Astronomy Consortium (NAC) Internship ($7200) *July 2019*

2018
NMSU Astronomy William Webber Voyager Graduate Scholarship ($7500) *August 2018*
National Astronomy Consortium (NAC) Internship ($7200) *May 2018*

SERVICE & LEADERSHIP

Panel Reviews
NASA ROSES research grant competition: (Exec. Secretary) Take notes during discussions, help facilitate voting and prepare final reviews. *dates confidential*

Observing Proposal Peer Reviews
Observatories: ALMA *dates confidential*

National Astronomy Consortium (NAC)
National Radio Astronomy Observatory, Socorro, NM
NAC Summer Lead: Host weekly meetings, peer-mentorship with NAC students. *May 2024-August 2024*

National Astronomy Consortium (NAC)
National Radio Astronomy Observatory, Socorro, NM
NAC Coordinator Assistant: Help develop programming for the 2021 cohort of NAC students. *April 2021-May 2021*

NMSU Graduate Workers United (NMSUGWU)
New Mexico State University, Las Cruces, NM

Social Media Manager: Manage Facebook, Instagram, and Twitter accounts for the NMSU Graduate Workers United. *August 2020-May 2021*

Astronomy Graduate Student Organization (AGSO)
New Mexico State University, Las Cruces, NM
Social Media Manager: Manage Instagram and Twitter accounts for the NMSU Astronomy Department. *August 2020-May 2021*

Astronomy Graduate Student Organization (AGSO)
New Mexico State University, Las Cruces, NM
Extracurricular Activities Organizer: Coordinate, plan, and inform fellow astronomy graduate students of extracurricular events. *August 2019-May 2020*

Society of Physics Students (SPS)
Old Dominion University, Norfolk, VA
Vice President: Assisted the President of SPS in planning and hosting events, meetings, etc. Responsible for organizing SPS group events at the Pretlow Planetarium. *May 2017-May 2018*

TELESCOPE PROPOSALS

Apache Point Observatory (APO)
Project: Long-slit spectroscopy of emission line nebulae. *350 hours (22 quarters/2020-2025)*
PI: **Kelly N. Sanderson**
Co-Is: Moire K. M. Prescott, Audrey F. Dijeau, Anna D. Kapińska

Hubble Space Telescope (HST)
Project: A Complete Sample of Ionization Echoes Around Fading AGN
Co-I, HST Cycle 32 Guest Observer (GO-17831), *24 primary orbits (2024)*
Co-I, HST Cycle 32 Bridge Program (GO-17930), *22 primary orbits (2024)*

Giant Metrewave Radio Telescope (GMRT)
Project: Constraining the History and Power of Jetted AGN Activity. *33 hours (2026B)*
PI: **Kelly N. Sanderson**
Co-I: Nirupam Roy

Very Large Array (VLA)
Project: History of Jet Production in Green Bean Galaxies. *5.8 hours (2026B)*
PI: **Kelly N. Sanderson**
Co-I: Anna D. Kapińska

Very Long Baseline Array (VLBA)
Project: VLBI imaging of an unusual radio galaxy structure. *1 hour (2021)*
PI: **Kelly N. Sanderson**
Co-I: Anna D. Kapińska

Project: VLBI astrometry of a binary star system. *33 hours (2018)*
PI: Anthony Beasley
Co-Is: Tim Bastian, **Kelly Sanderson**

PUBLICATIONS

Refereed: 6 **Non-refereed:** 1

7. **The Green Chile Project I: A New Catalog of Low-redshift Extended Emission Line Nebulae**
Audrey F. Dijeau, Moire K. M. Prescott, **Kelly N. Sanderson**
2025, AJ, 169, 207 10.3847/1538-3881/adb485

6. **SAUNAS II: Discovery of Cross-shaped X-ray Emission and a Rotating Circumnuclear Disk in the Supermassive S0 Galaxy NGC 5084**
Alejandro S. Borlaff, Pamela M. Marcum, Pasquale Temi, Nushkia Chamba, S. Drew Chojnowski, Enrique Lopez-Rodriguez, Aneta Siemiginowska, Seppo Laine, Anton M. Koekemoer, **Kelly N. Sanderson**, Audrey F. Dijeau, Moire K. M. Prescott, Leslie Proudfit, Michael N. Fanelli
2024, ApJ, 977, 238 10.3847/1538-4357/ad7c4b

5. **Signatures of AGN feedback modes: A Green Bean Galaxy with 150 kpc jet-induced radio emission**
Kelly N. Sanderson, Anna D. Kapińska, Moire K. M. Prescott, Audrey F. Dijeau, Savannah R. Gramze, Jacqueline Hernandez, Katherine T. Kauma
2024, ApJ 971, 20 10.3847/1538-4357/ad50a8

4. **Prevalence of a growth mindset among introductory astronomy students**
Moire K. M. Prescott, Laura Madson; Sandra M. Way, **Kelly N. Sanderson**
2024, PRPER 20, id.010140, 10.1103/PhysRevPhysEducRes.20.010140

3. **VLA Radio Detection of MAXI J1816-195**
Isaiah Beauchamp, Stefania Belvedere, Mitchell Hernandez, Morgan Himes, Trevor McCaffrey, **Kelly Sanderson**, and Bailee Wolfe
2022, The Astronomer's Telegram, No. 15481

2. **Mapping the Morphology and Kinematics of a Lyα-selected Nebula at z = 3.15 with MUSE**
Kelly N. Sanderson, Moire K. M. Prescott, Lise Christensen, Johan Fynbo, Palle Møller
2021, ApJ 923, 252 doi:10.3847/1538-4357/ac3077

1. **Understanding Lyman-Nebulae at Low Redshift I: The Sizes, Powering, and Kinematics of Green Bean Galaxies**
Moire K. M. Prescott and **Kelly N. Sanderson**
2019, ApJ 885, 40 doi:10.3847/1538-4357/ab44b1

PRESENTATIONS

21. **Constraining Supermassive Black Hole Activity Using Extended Emission Line Nebulae and Jet-induced Radio Emission**
Galaxy Zoo Telecon
Oral, April 24, 2026
Co-authors: Prescott, M.K.M., Christensen, L., Møller, P., Fynbo, J.P.U., Kapińska, A., Dijeau, A.F., Gramze, S.R., Hernandez, J., Kauma, K.T.

20. **Constraining Supermassive Black Hole Activity Using Extended Emission Line Nebulae and Jet-induced Radio Emission**
247th AAS Meeting
Oral, January 6, 2026
Co-authors: Prescott, M.K.M., Christensen, L., Møller, P., Fynbo, J.P.U., Kapińska, A., Dijeau, A.F., Gramze, S.R., Hernandez, J., Kauma, K.T.

19. **Constraining Supermassive Black Hole Activity Using Extended Emission Line Nebulae and Jet-induced Radio Emission**
41st New Mexico Symposium
Oral, November 21, 2025
Co-authors: Prescott, M.K.M., Christensen, L., Møller, P., Fynbo, J.P.U., Kapińska, A., Dijeau, A.F., Gramze, S.R., Hernandez, J., Kauma, K.T.

18. **Constraining Supermassive Black Hole Activity Using Extended Emission Line Nebulae and Jet-induced Radio Emission**

NMSU Dissertation Defense
Oral, March 31, 2025

17. **Exploring Active Galaxies with Extended Emission Line Nebulae and Jet Signatures**
NRAO WUNA Lunch Talk
Oral, December 4, 2024
Co-authors: Prescott, M.K.M., Christensen, L., Møller, P., Fynbo, J.P.U., Kapińska, A., Dijeau, A.F., Gramze, S.R., Hernandez, J., Kauma, K.T.

16. **Signatures of AGN Feedback Modes in a Green Bean Galaxy**
Annual NACtober Meeting (Alum Talk)
Oral, October 20, 2024
Co-authors: Prescott, M.K.M., Dijeau, A.F., Kapińska, A., Nava, N., Gramze, S.R., Hernandez, J., Kauma, K.T.

15. **Green Bean Galaxies: Results from the Prescott Research Group**
APO Staff Lunch Talk (Invited)
Oral, May 10, 2024
Co-authors: Prescott, M.K.M., Dijeau, A.F., Kapińska, A., Nava, N., Gramze, S.R., Hernandez, J., Kauma, K.T.

14. **A Multi-frequency Analysis of the Timescales Associated with AGN Activity in a Jetted Green Bean Galaxy**
AGN on the Beach: The many facets of fuelling and feedback in jetted AGN
Oral, September 13, 2023
Co-authors: Kapińska, A., Prescott, M.K.M., Dijeau, A.F., Gramze, S.R., Hernandez, J., Kauma, K.T.

13. **Mapping the Morphology and Kinematics of a Lya-selected Nebula at z=3.15 with MUSE**
Oases in the Cosmic Desert: Understanding the Structure of the Circumgalactic Medium
Poster + 1 min Lightning Talk, February 21, 2023
Co-authors: Prescott, M.K.M., Christensen, L., Møller, P., Fynbo, J.P.U.

12. **Mapping the Morphology and Kinematics of a Lya-selected Nebula at z=3.15 with MUSE**
VLASS in the Multiwavelength Spotlight
Poster, September 7-9, 2022
Co-authors: Prescott, M.K.M., Christensen, L., Møller, P., Fynbo, J.P.U.

11. **Investigating the Evolution of Radio Galaxies with Lyman alpha Nebulae Across Cosmic Time (Year 2)**
New Mexico Space Grant Consortium (NMSGC) Virtual Student Research Colloquium
Oral, April 21, 2022
Co-authors: Prescott, M.K.M., Kapińska, A.

10. **Mapping the Kinematics of the Cosmic Web at z∼3.2**
37th New Mexico Symposium
Oral, November 19, 2021
Co-authors: Prescott, M.K.M., Christensen, L., Møller, P., Fynbo, J.P.U.

9. **Radio Galaxies with Extended Gas Reservoirs at $0.2 < z < 0.6$**
2021 Virtual APO Science Symposium
Oral, July 28, 2021 Co-authors: Prescott, M.K.M., Kapińska, A.

8. **Investigating the Evolution of Radio Galaxies with Lyman alpha Nebulae Across Cosmic Time (Year 1)**

New Mexico Space Grant Consortium (NMSGC) Virtual Student Research Colloquium
oral, April 22, 2021
Co-authors: Prescott, M.K.M., Kapińska, A.

7. **Introduction to Galaxy Zoo**
Apache Point Observatory Open House
Evening lightning talk series
oral, March 7, 2020

6. **Mapping the Kinematics of the Cosmic Web at z∼3.2**
35th New Mexico Symposium
Oral, February 21, 2020
Co-authors: Prescott, M.K.M., Christensen, L., Møller, P., Fynbo, J.P.U.

5. **Mapping the Kinematics of the Cosmic Web at z∼3.2**
235th AAS Meeting
Oral, January 8, 2020 Co-authors: Prescott, M.K.M., Christensen, L., Møller, P., Fynbo, J.P.U.

4. **Investigating Radio Emission from a Green Bean Galaxy using the VLA**
NAC VII Meeting
Oral, September 14, 2019
Co-authors: Kapińska, A. D., Beasley, A.

3. **Investigating the Possible Mechanisms Powering the Radio Pulsations in AR Scorpii**
233rd AAS Meeting/NAC VI Meeting
Poster & Oral, January 8, 2019 Co-author: Beasley, A.

2. **The Evolution of Nuetral Hydrogen in Galaxy Groups**
231st AAS Meeting
Poster, January 9, 2018
Co-author: Wilcots, E. M.

1. **The Evolution of Nuetral Hydrogen in Galaxy Groups**
NAC V Meeting
Oral, September 16, 2017
Co-author: Wilcots, E. M.

MEDIA COVERAGE

AAS Journal Author Series: Kelly Sanderson on 2024ApJ...971...20S

SKILLS

Programming	Python, LaTeX, CASA, AIPS, SCHED, Git, SQL, C++
Interpersonal	Teamwork-oriented, Collaborative, Event Planning

OUTREACH & COMMUNITY SERVICE

Very Large Array (VLA) Open House
VLA, NM *October 12, 2023*
Provided visitors with information on the whisper dishes on-site at the VLA.

VLASS in the Multiwavelength Spotlight
Braodening Participation (BP) Panel co-chair *September 8, 2022*
New Mexico Institute of Mining and Technology (NMT)
Introduced speakers for the Broadening Participation (BP) panel and kept the flow of the session running smoothly.

Astronomy Club Guest Speaker
University High School, Irvine, California - Virtual *March 3, 2021*
Talked about galaxy evolution and my academic career.

Astronomy Youtube Channel
Kelly Sanderson *2020 - 2022*
Create astronomy coding tutorials, and videos about research and academia in astronomy.

NRAO NAC VII Guest Speaker
Virtual *July, 2020*
Met with NRAO NAC summer students to talk about how to read and summarize scientific papers.

STEM Explorers Camp
Virtual *June 22-26, 2020*
Met with elementary students for their virtual summer camp to talk about astronomy and do hands on activities.

Apache Point Observatory (APO) Open House
APO, Sunspot, NM *March 7, 2020*
Taught visitors about comet composition. Presented 5 minute talk introducing the Galaxy Zoo citizen science project.

Southern NM Pride Festival
Plaza de Las Cruces, NM *October 05, 2019*
Set up telescopes, taught visitors about meteorites, and answered public questions.

Apollo 11 50th Anniversary Star Party
Tombaugh Observatory, Las Cruces, NM *September 19, 2019*
Set up telescopes, assisted visitors in observing, and answered public questions.

Math & Science Night
Vado Elementary School, Las Cruces, NM *January 17, 2019*
Set up telescope and answered public questions.

Fall Festival
St. Paul's UMC, Las Cruces, NM *October 13, 2018*
Set up educational activities to teach participants about galaxy colors and comet composition.

International Observe the Moon Night
Norfolk, VA *September 19, 2015*
Helped with interactive activities to teach the public about the Moon.

26th Annual Children's Festival
Norfolk Harbor, VA *October 4, 2014*
Helped set up a table for the physics department. Demonstrated experiments for the public.